DO NOW — RECALL

OCR pseudocode, files and IDEs

Name: _____ Class: _____ Date: _____

1. If y=4 is passed by ref and a method adds 5, what is y after the call?

2. What does a breakpoint do?

3. Which condition controls a while loop?

4. If a file has two lines and each readLine reads one line, how many calls read the whole file?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

9.

2. Model answer

Pauses execution at a specified line.

3. Model answer

A Boolean test evaluated before each iteration.

4. Model answer

Two calls, assuming both lines are read and no additional data.

Vocabulary — know these terms

openRead — Open a file for reading using OCR A Level pseudocode.

readLine — Read the next line from an open file.

endOfFile — Test whether no unread lines remain.

openWrite — Open for writing; existing file contents are replaced.

close — Release an opened file resource.

NEW LEARNING

Read, annotate and explain

Open the resource

openRead returns a file object for reading; it does not overwrite the file.

Read until EOF

endOfFile() is false while another line remains. readLine() retrieves one line and moves the file position.

Close the file

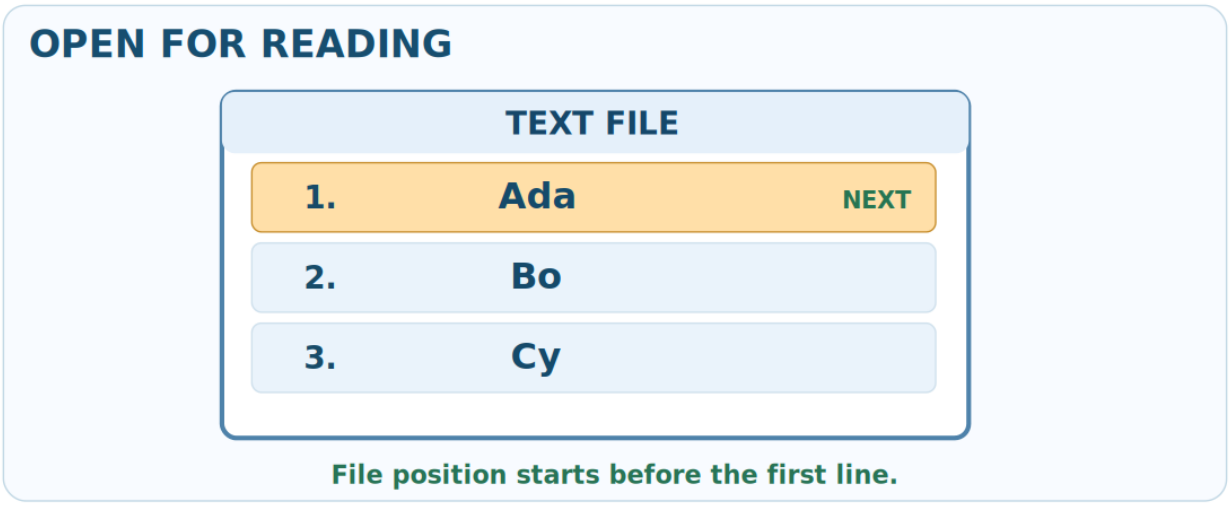
After the final line, endOfFile() is true, so the loop stops and the file is closed. For writing use openWrite , writeLine , then close() ; unlike reading, opening for writing overwrites any existing file contents .

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

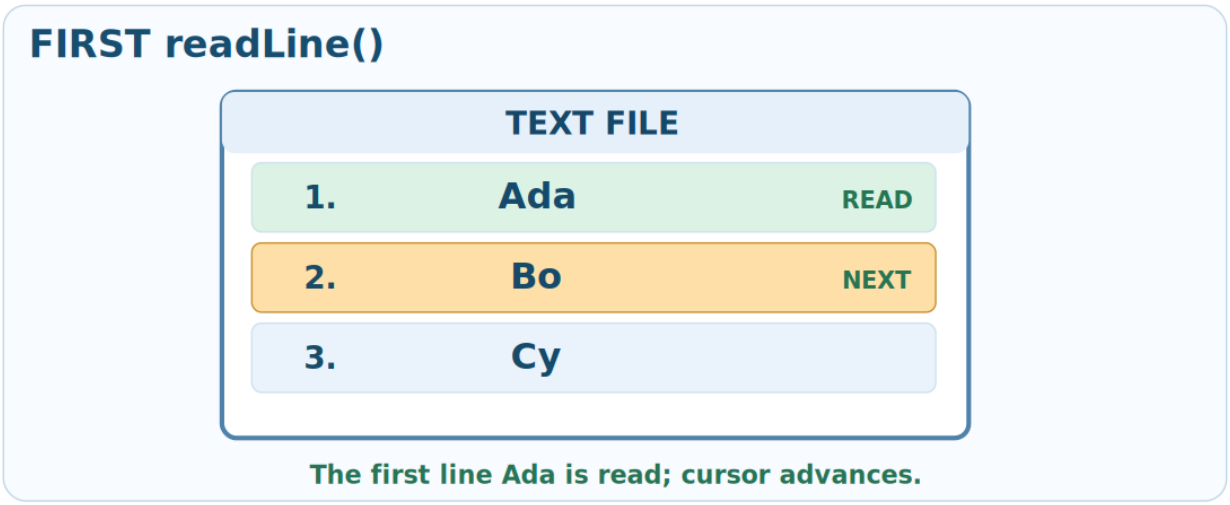
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



OPEN FOR READING: File position starts before the first line.

New Learning

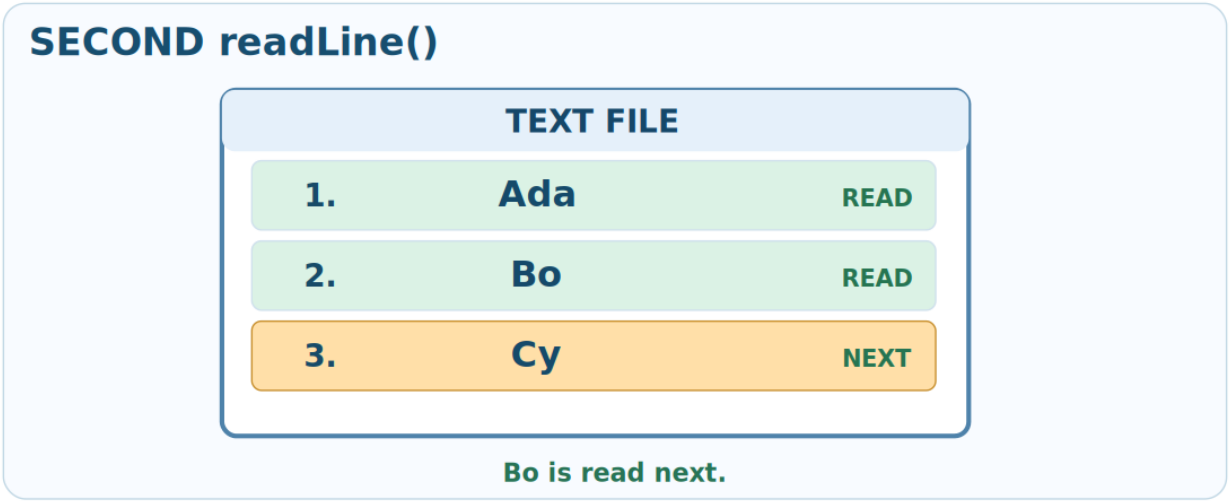


FIRST readLine(): The first line Ada is read; cursor advances.

New Learning · original lesson diagrams

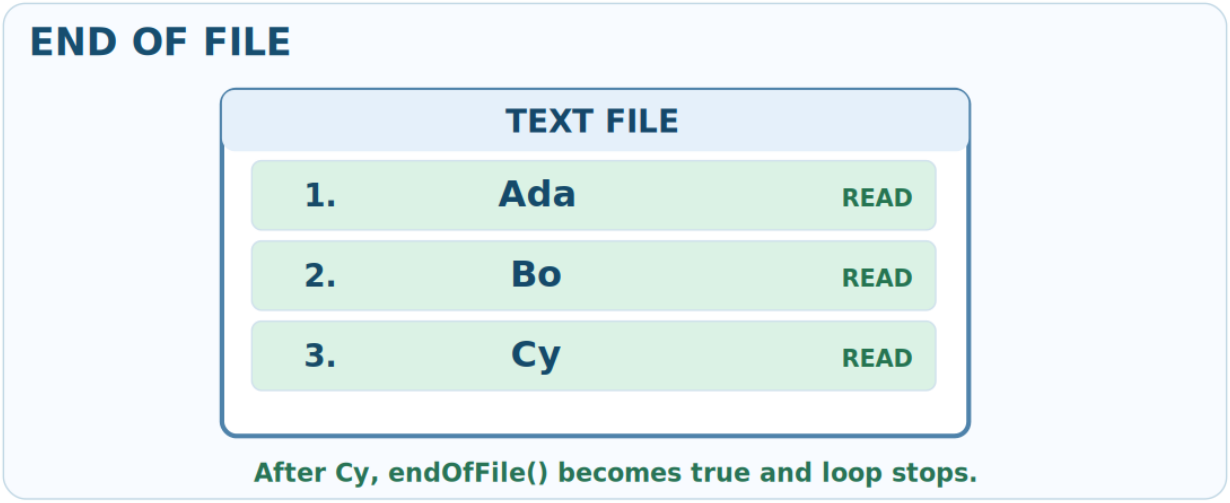
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



SECOND readLine(): Bo is read next.

New Learning



END OF FILE: After Cy, endOfFile() becomes true and loop stops.

I DO — WORKED EXAMPLE

Follow the worked method

Trace OCR pseudocode that reads the first two lines of fruits.txt.

Predict first

The first call returns Apple and advances the file position.

Model and check

The next call returns Pear; variable a stays Apple.

Model and check

close releases the file resource; Plum has not been read.

Write mode is not read mode

openWrite opens the named text file for writing, replacing any previous contents. Do not use it simply to inspect scores already stored there.

Two calls create two lines

After the first call the file contains Ada,82 . After the second it also contains Ben,74 on the next line. A text file is not automatically a database table.

Predict a later read

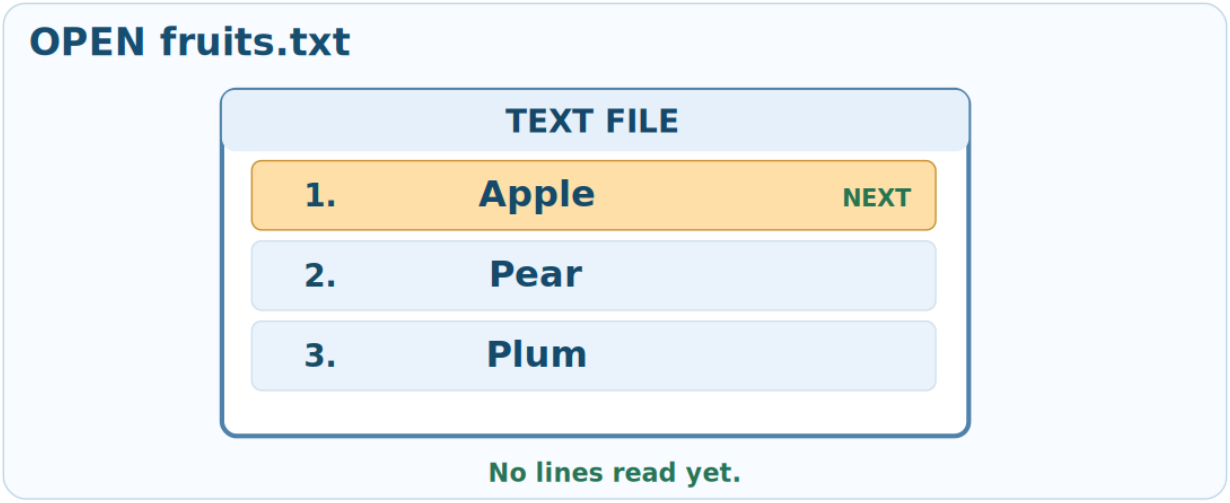
f.close() releases the file. A separate later openRead("scores.txt") would read Ada,82 then Ben,74 in that order.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace OCR pseudocode that reads the first two lines of fruits.txt.

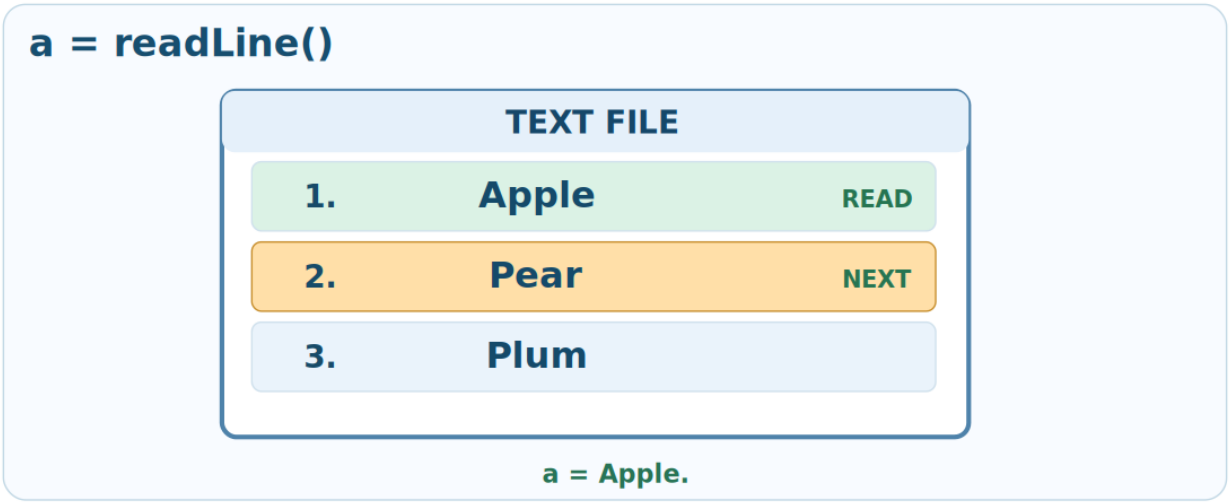
The first call returns Apple and advances the file position.



OPEN fruits.txt: No lines read yet.

Trace OCR pseudocode that reads the first two lines of fruits.txt.

The first call returns Apple and advances the file position.



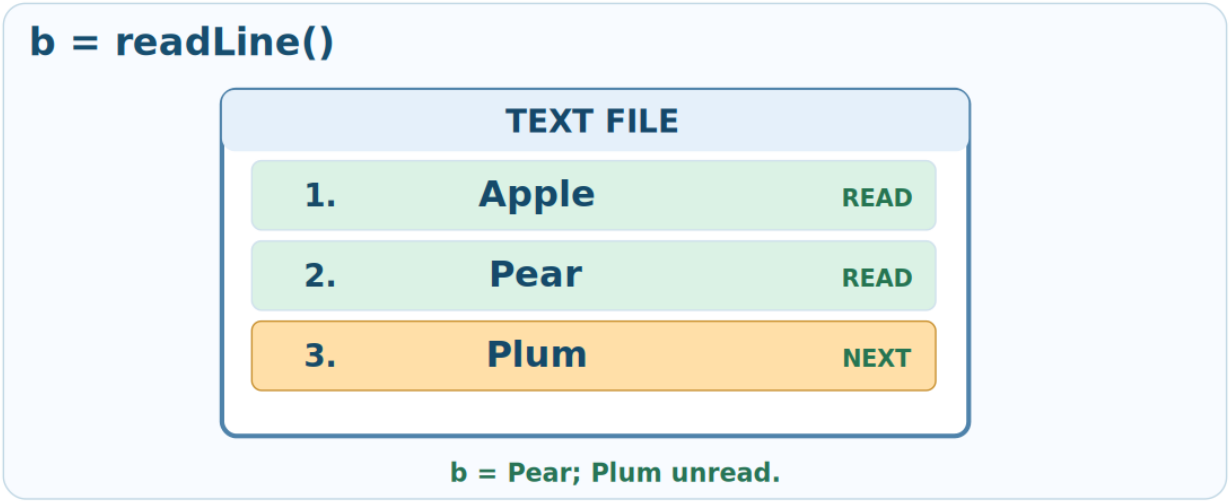
a = readLine(): a = Apple.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace OCR pseudocode that reads the first two lines of fruits.txt.

The first call returns Apple and advances the file position.



b = readLine(): b = Pear; Plum unread.

WE DO — GUIDED PRACTICE

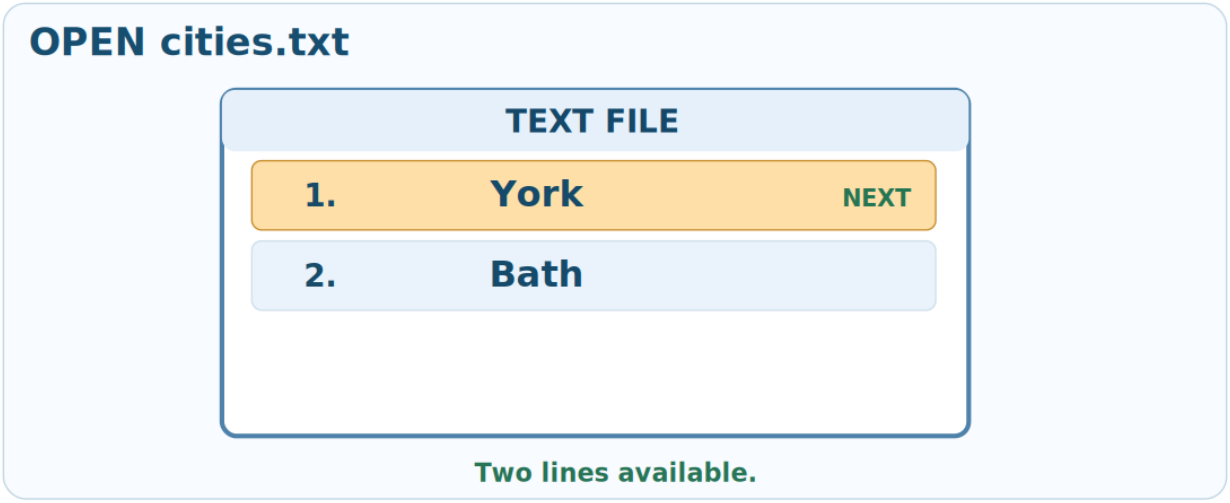
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a loop reading all lines of cities.txt. How many readLine calls occur?

The loop prints York from the first readLine call.



OPEN cities.txt: Two lines available.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first The loop prints York from the first readLine call.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

An OCR pseudocode file contains Cat, Dog, Fox on separate lines. What do the first two readLine() calls return?

Question 2

Write OCR A Level pseudocode to open words.txt, display each line and close the file.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why openWrite is unsuitable for reading an existing file. Write OCR pseudocode that replaces events.txt with one line HELLO, and identify a debugging step if a reading loop omits its last record.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

OCR's openWrite("data.txt") safely reads existing data without altering it.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

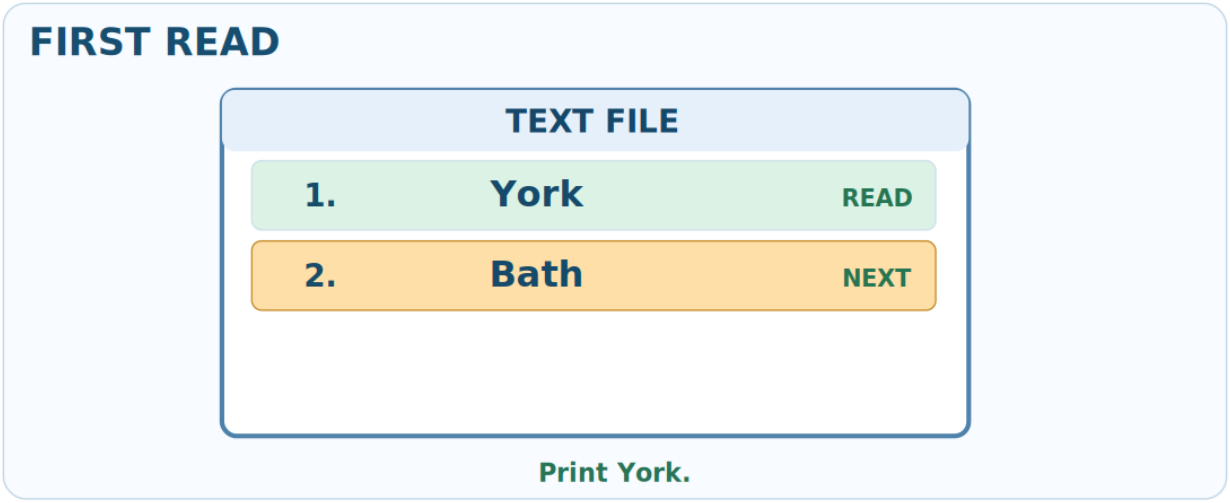
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a loop reading all lines of cities.txt. How many readLine calls occur?

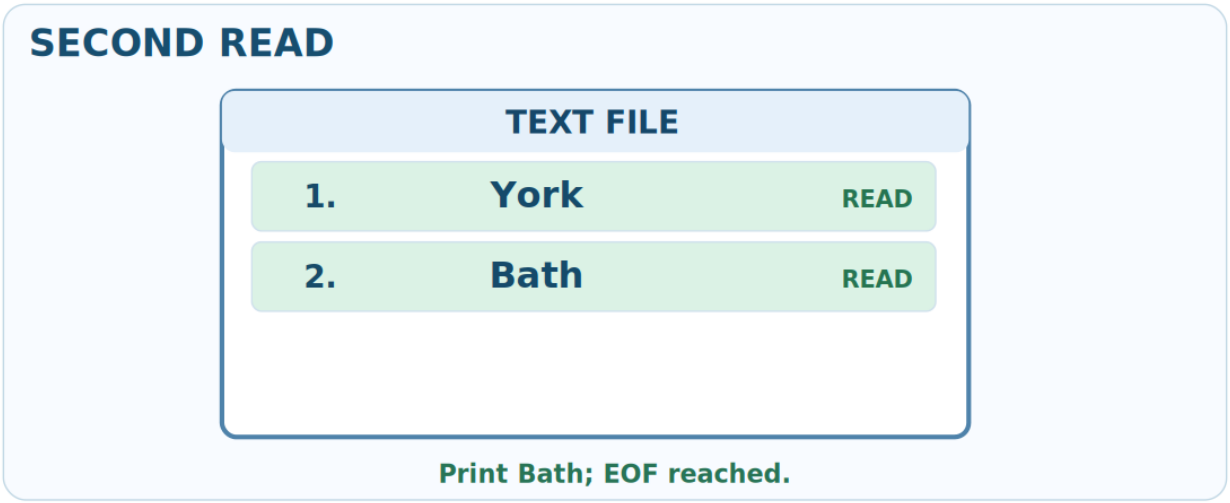
The loop prints York from the first readLine call.



FIRST READ: Print York.

Trace a loop reading all lines of cities.txt. How many readLine calls occur?

The loop prints York from the first readLine call.



SECOND READ: Print Bath; EOF reached.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and check

Then it prints Bath from the second call. At EOF the loop condition becomes false, so it stops after two reads.

IMPROVE: Explain one correction to your own reasoning.